

**How is vehicle  
dominance made?**



**Centre for Blockchain  
Technologies**

# **The Making of Dominant Vehicle Currencies**

Evidence from Ethereum routes  
Jiahua Xu

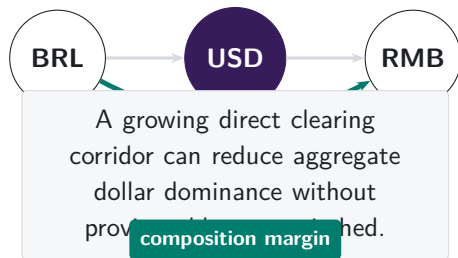
# Dominance changes when trade corridors change

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## TRADITIONAL-FINANCE INTUITION

A dollar share can fall even if old dollar corridors stay dollar-settled.

Growth in a non-dollar corridor changes the aggregate denominator. That is a route-allocation margin, not only a within-pair substitution margin.



*Note:* The clearing example is an analogy for the denominator problem. The empirical object in this paper is the intermediary asset observed inside DEX pool routes.

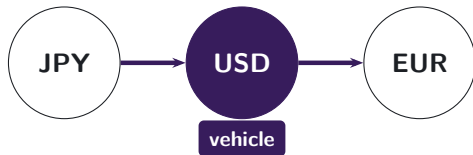
# Vehicle currencies are routes

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## FX ANALOGY

Yen–euro may trade through dollars.

Pair data see two legs, not the customer route.



Pool logs reveal the path.

DEX data make the hidden route observable.

*Note:* Traditional pair-level FX records show turnover in each bilateral pair, but not whether two legs belong to one customer route. The DEX setting preserves the transaction-level pool path, so the intermediary asset is observed directly.

# The PYUSD swap reaches a USDC-funded pool route

## USER INSTRUCTION RECORDED BY THE EXPLORER

**Transaction details** # 0x

⚡ Swap 460.00K  PYUSD for 460.10K  USDE on 0x  0x

## THE DECISIVE SETTLEMENT TRANSFERS

 USDC

ERC-20 **Transfer**

 Market Maker: 0x51c7.....  →  Mainnetsettler 

 USDC

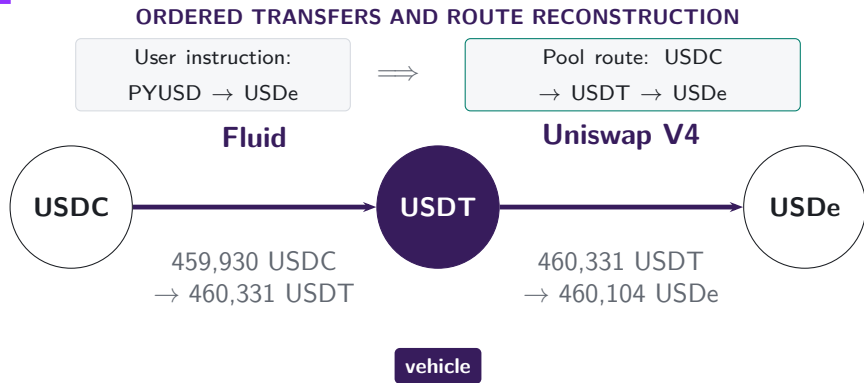
ERC-20 **Transfer**

 Mainnetsettler  →  Fluid: Liquidity 

The instruction is PYUSD → USDe. The maker supplies the USDC used by the Fluid leg, which begins the connected pool route.

*Note:* The explorer screenshot and transfer trace refer to transaction 0xbda1d07f...9bf67ad9 on 10 January 2026.

# Inside the pools, USDT links USDC to USDe



The observed ultimate trade is USDC → USDe. Its atomic trades are USDC → USDT and USDT → USDe; USDT is the vehicle that links them.

*Note:* The route is the connected pool-swap component of transaction 0xbda1d07f...9bf67ad9 on 10 January 2026.

# Vehicle dominance is the share of indirect trade carried by an asset

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## MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset  $k$ .
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly  $k$  connects ultimate pairs.



### Count-weighted dominance

Share of indirect routes that pass through  $k$

### Value-weighted dominance

Share of comparable routed value that passes through  $k$

# Architecture changes routing opportunities

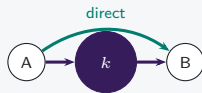
## V1 mandatory hub



Only ETH-token pools: token-to-token execution must pass through ETH.

**Hub use is imposed**

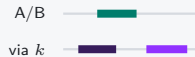
## V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

**Hub use becomes a choice**

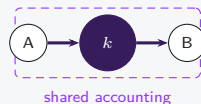
## V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

**Opportunity shifts by pair**

## V4 net settlement



Two pools still execute; intermediate token movement may be netted.

**Settlement can diverge from route use**

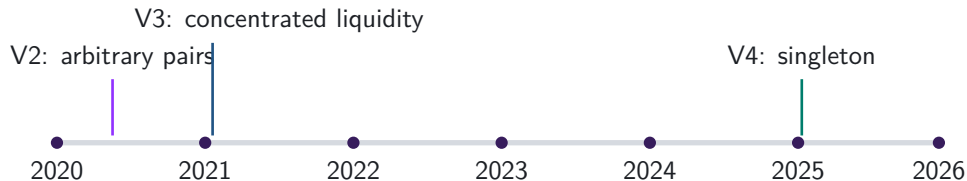
**Architecture** feasible rules

**Exposure** realised pool and route use

**Calendar date** timing only

# The route sample and architecture supplement answer different questions

## DATA



## Main route sample

Atomic trades (swap legs) from multiple protocols are linked within each transaction. An atomic pair may be a two-asset pool or one asset pair inside a multi-asset Curve or Balancer pool.

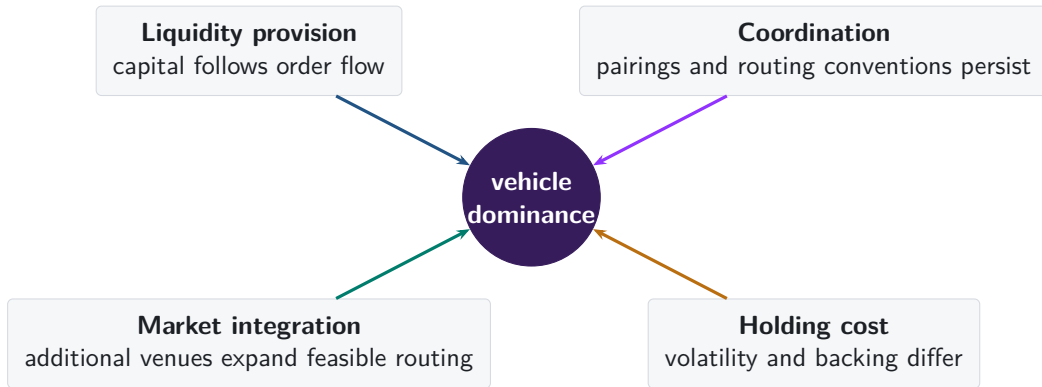
## V1/V2 architecture supplement

Uniswap V1 forced-ETH paths and V2 pair formation are analysed separately. They answer protocol-design questions and do not define the main route perimeter.



# Why might one asset become the dominant vehicle?

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# Native-asset pairing persists after it becomes optional

## V1: PROTOCOL RULE



**217,003 token-to-token trades  
pass through ETH**

Every token pool pairs with ETH, so the architecture imposes the vehicle.

## V2: MARKET CHOICE

**95.5%**

of single-leg V2 trades use a WETH pool in 2026

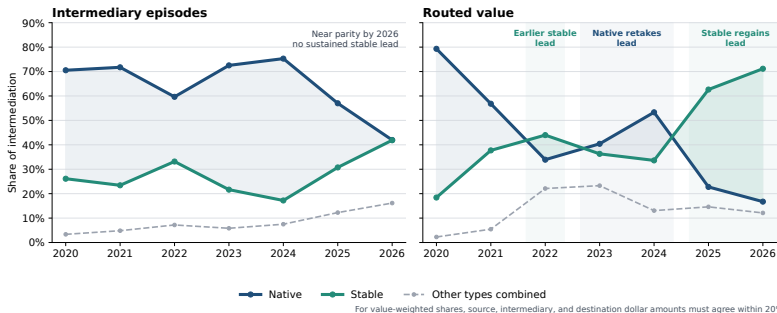
**97.9%**

of pairs first traded in 2026 include WETH

V2 converted ETH pairing from a protocol rule into a market choice. Native-asset concentration remained high through pair formation and realised trading.

*Note: V1 counts linked token-to-ETH and ETH-to-token swaps with matching ETH quantities in one transaction. V2 shares use single-leg Uniswap V2 trades and newly traded pairs; they describe that venue rather than market-wide vehicle dominance.*

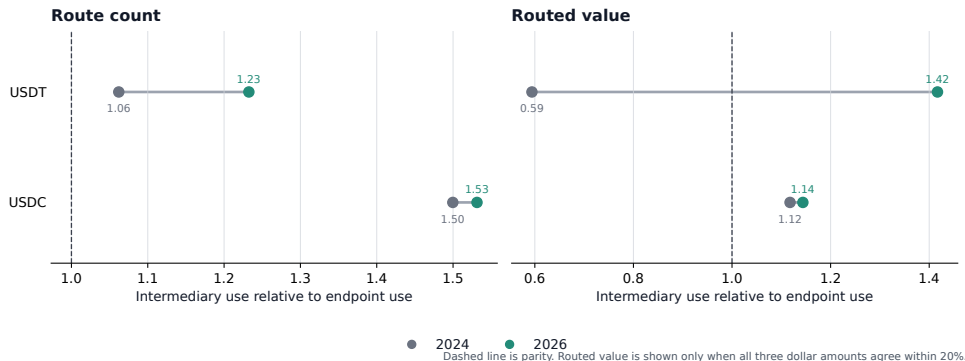
# Stablecoins regain the lead in routed value



The aggregate share moves because the network of traded ultimate pairs reorganises around stablecoins. Holding the ultimate pair, the calendar date, and the route scope fixed, stablecoin share changes by +0.2 pp (SE 0.8 pp) by count and -1.3 pp (SE 2.2 pp) by value.

*Note:* The route universe is exact two-leg routes (two atomic trades) with an observed intermediary. The left panel counts one intermediary episode per route, hence per ultimate trade, and divides each asset type's episodes by all intermediary episodes. The right panel divides each type's routed value by total routed value among routes whose source, intermediary, and destination dollar amounts agree within 20%. The 2026 observation covers January–June; Other combines imported, liquid-staking, and residual assets. Matched-market estimates weight by route count or value and cluster by ordered ultimate pair and calendar date.

# USDT drives the rise in stablecoin intermediation



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

*Note:* Excess use is intermediary share divided by endpoint share on the same route perimeter. The comparison uses native assets, USDC, and USDT and the same January–June dates in 2024 and 2026. Descriptive accounting: see Appendix A9; the identity carries no sampling interval.

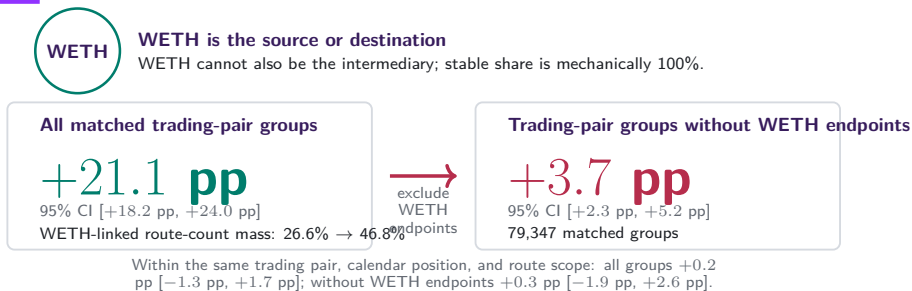
## The cross-section holds when the calendar is absorbed

|                    | Native    | Stable   | Demand |
|--------------------|-----------|----------|--------|
| Pooled             | +34.56 pp | +2.42 pp | —      |
| + day effects      | +34.55 pp | +2.42 pp | —      |
| + own demand       | +0.05 pp  | +0.85 pp | 1.587  |
| + currency effects | —         | —        | 0.985  |

Day effects leave the class premiums where they were. At the same trade demand the native premium is +0.05 pp (0.56 pp) and the stablecoin premium +0.85 pp (0.50 pp).

*Note:* 1,828,862 currency-days, 304,572 currencies, 2,259 days. Premiums are percentage points of the day's intermediary episodes against the residual unclassified bucket; demand is the pass-through of the currency's own route-endpoint share. Standard errors cluster on date and currency.

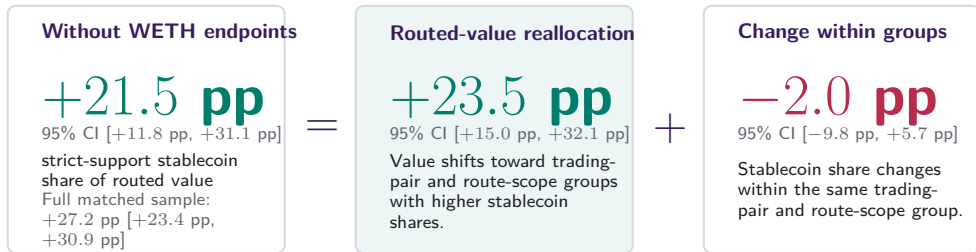
# WETH-linked ultimate pairs account for most of the count rotation



The count increase falls from +21.1 pp to +3.7 pp without WETH endpoints. A WETH-linked pair admits only a stablecoin intermediary in this comparison, so the count rotation largely records where trading happens rather than which intermediary a given trade selects.

*Note:* Exact two-leg native-WETH-versus-stablecoin routes on 181 common month-days. Intervals use 30-day Newey–West covariance on 362 endpoint-year daily observations; 2025 is excluded and actual dates preserve the gap. Within-group WLS fixes the ultimate pair, month-day, and route scope and clusters by pair and date. Feasible routes are not fixed; all comparisons are descriptive.

# The routed-value rotation extends beyond WETH-linked ultimate pairs



Exact midpoint identity on each of 181 common calendar days; 2024 versus January–June 2026.

After removing WETH endpoints, value moves toward trading-pair and route-scope groups in which stablecoins already carry more intermediation. The change within those groups is small and imprecise.

*Note:* Routed value requires source, intermediary, and destination dollar amounts to agree within 20%. The daily midpoint identity is exact. Intervals use the same endpoint-year calendar-HAC covariance as the preceding frame; paired-month-day and bandwidth sensitivities remain in the source. Mukhin (2022), Gopinath and Stein (2021), and Carletti et al. (2021) guide the composition framing, not identification.

# Three margins move an aggregate vehicle share

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## 1 Used more often within a pair

USDC → WBTC trades in both years

Stablecoin share of its intermediary use

**10.5% → 35.9%**

Adds +0.05 pp to the aggregate

**Margin total: -0.1 pp**

## 2 Trading moves toward pairs that already use it

USDT → WETH, stablecoin-routed in both years

Share of all routed activity

**0.36% → 4.52%**

7,447 routes become 54,112; adds +2.17 pp

**Margin total: +8.6 pp**

## 3 Newly traded pairs route through it

USDS → WETH, absent from the base year

Routes observed

**0 → 3,390**

All stablecoin-intermediated; adds +0.13 pp

**Margin total: +21.0 pp**

The first margin is the one vehicle-currency theory describes, and it did not move. The third nets to +17.8 pp because 48.0% of routes run on pairs alive in only one year.

*Note:* Pooled route-count share among native and stablecoin intermediaries, 2024 against 2026 on 181 common calendar days. Each panel names the largest contributing ordered pair with two canonical symbols; unlabelled assets carry most of the third margin. Contributions are an exact allocation of the aggregate change and are descriptive.



## Vehicle choice is sticky from market birth

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**5.0%**

stable share at entry,  
2024

**25.1%**

stable share at entry,  
2026

**+97.0 pp**

30-day stable-born gap

Entrant state sticks: 2026 stable-born pairs stay at 97.5%, native-born at 0.5%. With controls, +10 pp entry stable share predicts +7.4 pp at 120 days. Stable endpoints route 17.5% through stablecoins, versus 5.2% otherwise.

*Note:* Entrants are ordered pairs first observed in the matched January-June endpoint window with positive native-plus-stable route mass. WETH endpoints carry 18.9% of 2026 entrant mass. Path screen uses 599,870 non-WETH entrants, entry size, year, endpoint, direct-share, and complex-share controls. Stable-endpoint gap expands by +10.3 pp. Descriptive association only.

## USDC kept the vehicle role while stable capital stepped back

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**67.9%**

USDC stable-route share  
during stress

**+11.4 pp**

stable basket route-share  
change

**−2.0 pp**

stable basket capital-share  
change

Routes did not flee USDC during the SVB weekend: its stable-vehicle share moved −0.2 pp while stable-route activity rose 73.6%. LP capital moved the other way: post-window WETH capital share rose +3.1 pp and USDC capital share fell −3.2 pp.

*Note:* Daily route counts and V2 deposited-capital shares from the five-candidate route-capital panel. Windows are the 30 calendar days before March 10, the March 10–13 stress interval, and the 30 calendar days after. Route identity is within stable candidates; route and capital changes are within the five-candidate set. The event mixes peg stress, arbitrage, and demand shocks, so this is descriptive, not a treatment effect.

## Stable vehicles turn on at the market edge

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−2.7 pp

stable turn-on per log  
baseline routes

+27.1 pp

stable turn-on per baseline  
direct-share unit

−1.1 pp

stable-majority switch per log  
baseline routes

Thin/direct/complex corridors turn on. In mixed risk sets, reach predicts share. Stable remains negative, but 2026 USDT gets a +88.2 pp issuer premium after reach controls.

*Note:* Transition screen: pair-day-scope rows, month-day FE, harmonic weights, two-way CR1. Risk-set rows are observed native-stable candidate sets, not feasible alternatives; reach is leave-one same-day or prior-30-day pair-scope count. Issuer rows add USDC/USDT indicators to the same design. Market-size rows are descriptive; driver screen only.

## Native-only markets later add stable routes

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2.0%

30-day stable turn-on,  
bottom size bin

24.5%

30-day stable turn-on,  
top size bin

41.4%

30-day stable turn-on,  
oldest bin

Among active native-only pair-days, stable-vehicle use over the next month is much more common in thick and older markets. Month-day FE slopes stay positive for market thickness (+3.5 pp) and pair age (+1.6 pp).

*Note:* Rows are 1,690,478 active ordered pair-days across 620,849 pairs with positive native-or-stable vehicle use and zero stable vehicle use on the origin day. Outcome is any stable-vehicle use over the next complete 30 days. Quantile bins and FE estimates are weighted by origin market-route count; FE absorbs month-day and clusters by ordered pair and date. Descriptive hazard, not causal timing.

## Extra-hop execution became cheap enough

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**152.7%**

more gas than direct,  
stable-vehicle median in 2026

**\$0.08**

fixed extra-hop toll  
at 2026 medians

**\$762**

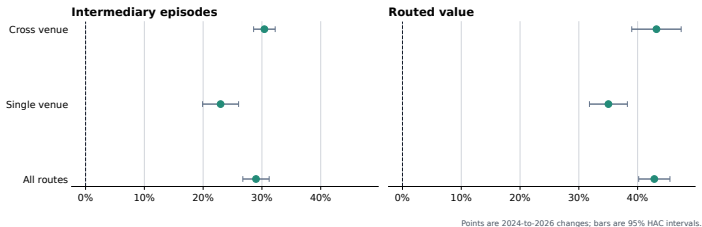
notional where extra gas is 1  
bp

In gas units, stable-vehicle routes still cost more than direct routes: the 2026 median extra toll is 233,640 gas. But median stable-vehicle notional fell from \$832 to \$82, while the median fixed toll fell from 44.3 bp to 16.2 bp.

*Note:* Executed-route receipt gas only. Dollar tolls evaluate the stable-vehicle extra-gas median against the same year's median gas price and WETH price. At route-specific gas prices, the share of stable-vehicle routes with fixed extra-hop toll below 25 bps rises from 39.5% to 55.7%; the 2026 ten-basis-point threshold is \$76, close to the observed stable-vehicle median notional of \$82. This is execution-friction evidence, not an all-in quote comparison.

# Stablecoin use rises within each venue scope

## Stable share rises within realised route scopes



Neither venue mix nor path length absorbs the rotation: two-leg routes gain +25.4 pp (SE 1.05 pp), longer routes +16.3 pp (SE 1.86 pp), and the weakest venue-by-length cell still rises by +17.0 pp. Nor does the pricing rule or automated path search. Within a day the venue gap is a value phenomenon: cross-exchange one-intermediary routes carry +14.18 pp more stablecoin value share (SE 3.69 pp), against +3.73 pp on  $[-2.2, +9.7]$  by count.

*Note:* Changes compare the same January–June dates in 2024 and 2026; bars show 95% calendar-HAC intervals. Excess-use ratios and the within-day venue gap span the full sample calendar.

## Routes follow local bridges, not only popular vehicles

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**84.1%**

route share captured by  
deepest bridge

**75.5%**  
**86.5%**

top-bridge share, 2024 to  
2026



**+6.74 pp**

route-share slope after  
candidate-reach controls

Deepest prior bridge carries most route mass. After reach controls, weak-leg depth remains +6.57 pp and leg imbalance predicts -5.42 pp. In stable-only cells, 2026 USDT adds +55.38 pp versus DAI.

*Note:* Five candidates; 137,003 candidate rows, 2,923 pairs, 363 days. Depth is the smaller prior-capital bridge leg. Bottleneck rows add stronger-leg or leg-imbalance controls. Reach uses same-day and prior-30-day routes. Stable-issuer rows use 12,048 DAI/USDC/USDT cells. Association only.

## Stable vehicle markets show feedback

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+0.115

later stable bridge-depth growth from a 10 pp higher current route share, 120 days

+2.46 pp

later stable route-share growth per log point of current local depth, 120 days

Route use predicts later bridge deepening, and depth predicts later route-share gains, on the same local source-candidate-target bridge. The feedback margin is stronger for stable candidates than for the pooled five-candidate set.

*Note:* Provisional feedback screen over 18,304 120-day rows and 340 ordered pairs. The stable 120-day log-depth coefficient is a 10 percentage point route-share effect, with SE 0.012. The stable 120-day route-share coefficient has SE 0.29 pp. Regressions control for current depth or current route share, absorb candidate and origin-date effects, weight by five-candidate route count, and cluster by ordered pair and date. Association only, not provider-flow causality.



## Provider capital stays native-heavy

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100.0%

WETH leads deposited capital

81.3%

stablecoins lead excess use

4.2×

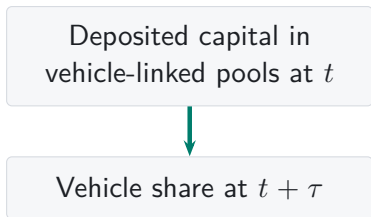
2026 route share over capital share

Stable route share rises while capital share falls. A 10 pp stable shortfall moves capital rank +0.23 by 120 days and incumbent-pool capital rises +12.39%, but same-pool fee yield moves only −0.27%. USDT V3 actions rise, provider-days +91.94%, net balance −0.66 pp.

*Note:* Five-candidate V2 capital panel. Rank rows order candidates within origin day and exact horizon; positive means nearer top share. Pool split is candidate/date FE over exact-state V2 pool capital. V3: 11,195 candidate-date mint/burn rows; origin counts are provider-days, not unique LPs; fee-yield uses processed fees over TVL.

# Deposited capital neither leads vehicle use nor follows it

## LIQUIDITY FIRST

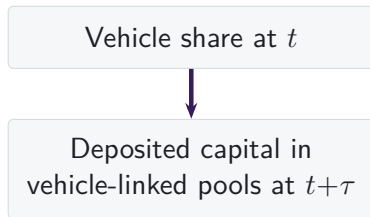


A one log-point higher capital stock predicts a next-day episode-share change of 0.00 pp (SE 0.04 pp); at thirty days  $-0.64$  pp (SE 0.26 pp).

No use/capital measure pair is positive in both directions at any two of the one-, seven-, and thirty-day horizons; the smallest Holm-adjusted  $p$  is 0.17. The only strong longer-horizon pattern runs the other way: over the following 120 days, higher-capital pool-days lose episode share ( $-2.21$  pp per log point, SE 0.39 pp).

*Note:* Daily panel of the five vehicle candidates on the full 2,238-day V2 calendar with candidate and date fixed effects; standard errors use 30-day calendar HAC on date-aggregated scores, and Holm adjustment is within direction; a month-block bootstrap and candidate-date clustering agree. Deposited capital is a reserve stock; provider flows and concentrated-liquidity depth are measured separately. Predictive associations only, not causal feedback.

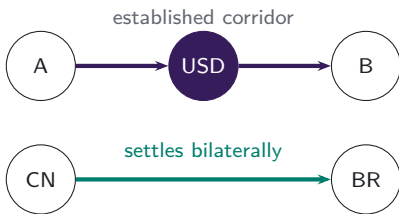
## VEHICLE USE FIRST



A one-point higher episode share predicts a next-day capital change of  $-0.03\%$  (SE 0.03%); at thirty days  $-0.32\%$  (SE 0.27%).

# The same arithmetic governs any aggregate currency share

## A FAMILIAR SETTING



China and Brazil have built clearing arrangements for settling their bilateral trade in renminbi. The established corridor above is unaffected.

## THE SHARE CAN FALL WITH NO CORRIDOR SWITCHING

A currency's share of all settled trade falls when trade grows fastest along corridors that never adopted it, even if every corridor that uses it today keeps using it tomorrow.

**Margin 2** is trade reallocating toward corridors that already avoid the incumbent.

**Margin 3** is a corridor that had no meaningful volume opening on other terms.

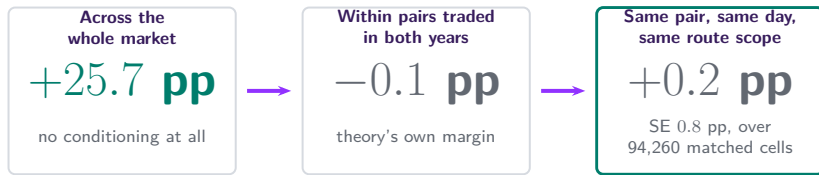
Our routes measure the same two margins directly, because the transaction log records the whole path rather than its legs.

*Note:* Illustrative comparison only. Renminbi clearing arrangements in Brazil follow the memorandum of understanding between the People's Bank of China and the Banco Central do Brasil. No currency share of trade settlement is measured here, and no causal or policy claim is made about either setting.

# The market changed, not the trade

## SHARE OF INTERMEDIATION CARRIED BY STABLECOINS

as the comparison holds more of the market fixed



**Stablecoins won through the markets that formed and grew, not repeated switching inside established pairs.**

measure at three levels of conditioning and are directly comparable. They describe realised route composition; they are not entry or exit effects and do not identify a treatment.

The three shares are one

# Appendix map

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## Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

## Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

## Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Observed route endpoints

## Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

# A1. One reconstructed route is one unit

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**one route unit**  
two swap legs

- Each linked  $i \rightarrow k \rightarrow o$  route is counted once.
- Asset  $k$  is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by  $k$ .

# How we reconstruct a route from one transaction

## Direct swap

one transaction



## Sequential vehicle route

one transaction



## Parallel split

one transaction



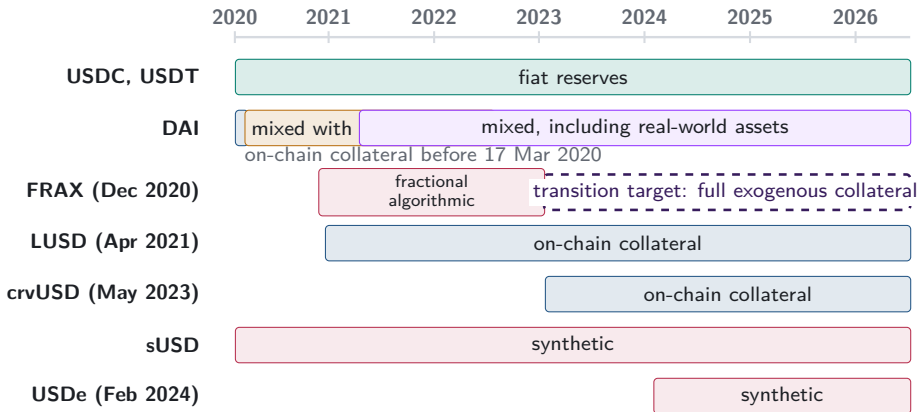
## Round trip

one transaction



Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths reveal that choice rather than leaving it inside aggregate turnover.

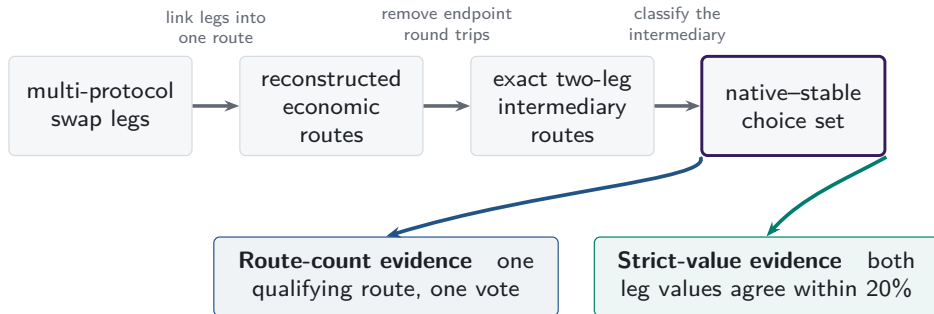
## A2. Stablecoin backing cannot be treated as fixed



*Note:* Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.



## A3. One route universe supports two measurements

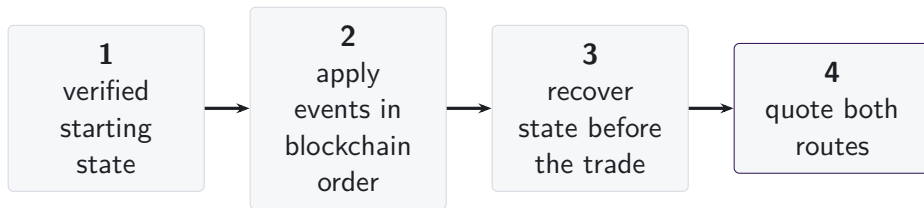


Box widths do not represent attrition.

The route sample links legs across protocols; one asset pair inside a multi-asset Curve or Balancer pool may supply a leg. The V1/V2 architecture evidence is separate. Dollar support affects value weighting, not route inclusion.

## A4. State is reconstructed immediately before execution

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- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

## A5. Each AMM family has a different state vector

|                        | reserves or<br>active liquidity | ticks | amplification | weights or<br>scaling | rates or<br>oracles | fees | hooks |
|------------------------|---------------------------------|-------|---------------|-----------------------|---------------------|------|-------|
| Constant product       | ■                               | —     | —             | —                     | —                   | ■    | —     |
| Concentrated liquidity | ■                               | ■     | —             | —                     | —                   | ■    | —     |
| StableSwap families    | ■                               | —     | ■             | —                     | ■                   | ■    | —     |
| Weighted pools         | ■                               | —     | —             | ■                     | ■                   | ■    | —     |
| V4 singleton           | ■                               | ■     | —             | —                     | —                   | ■    | □     |

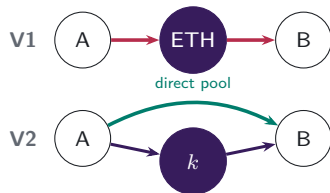
The common object is the state immediately before execution.  
The fields required to recover that state vary with the pricing rule.

■ required state

□ hooks outside comparison

# Protocol design changes markets, depth, and settlement

## V1 → V2: PAIR CREATION



V1 admits only ETH-token pools. V2 permits any ERC-20 pair, so market creation determines whether direct and vehicle paths coexist.

Which edges exist?

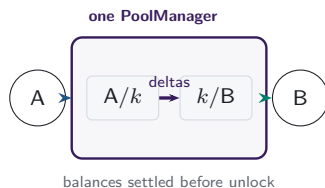
## V3: ACTIVE LIQUIDITY



LPs choose a price range and fee tier for each pool. Positions covering the current price determine active liquidity in the direct pair and in each vehicle spoke.

Which path offers depth at this size?

## V4: SHARED ACCOUNTING



One PoolManager holds all pools. During a lock, the two swaps update balance deltas; the caller settles its obligations before the lock ends.

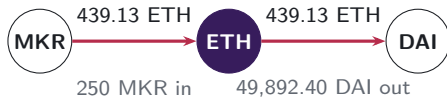
Where is the route settled?

**Economic implication.** Pair creation changes available paths; concentrated liquidity changes pair-specific execution at the current state and trade size; the singleton changes the settlement boundary for a route that still uses two pools.

## A6. V1 forced routes have an on-chain signature

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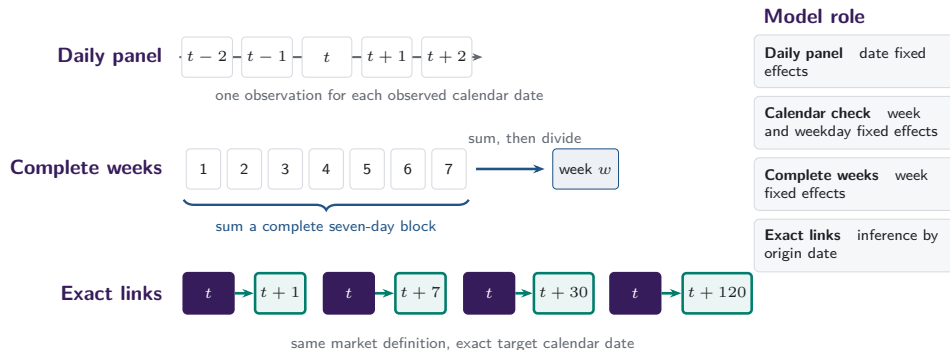
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



**same transaction, matching ETH flow**

*Note:* Both legs of transaction 0x4dca160d...96a0ca16 (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

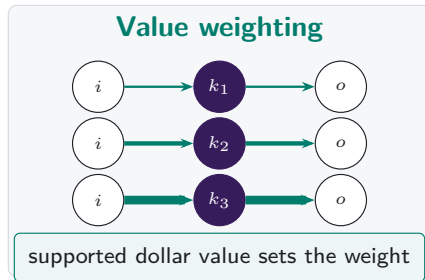
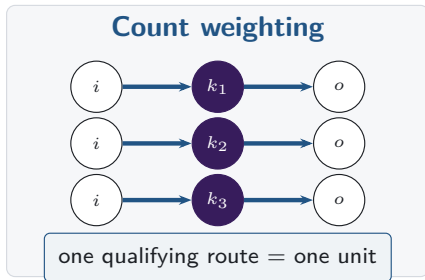
## A7. Daily and weekly frequencies answer different questions



Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

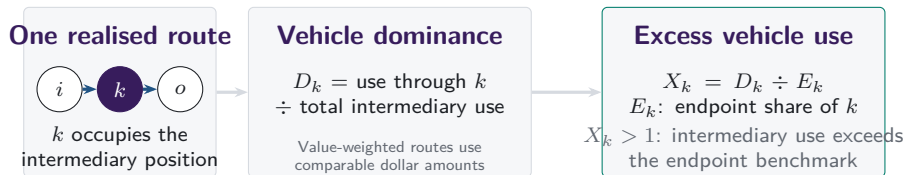
## A8. Count versus value weighting

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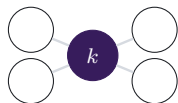


*Note:* Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

# Vehicle dominance aggregates realised intermediary choices

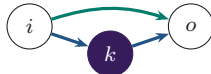


Separate characteristics of the asset and route set



## Network position

position of  $k$  in the feasible graph



## Execution cost

which route returns more at the same state

Observed route-endpoint use supplies the benchmark; network position and same-state route costs remain separate characteristics.



## A9. Intermediary and route-endpoint use are separate

---

### Intermediary share

$$I_{k,t}^N = \frac{n_{k,t}^I}{\sum_j n_{j,t}^I}$$

How often asset  $k$  sits inside a route.

### Endpoint share

$$E_{k,t}^N = \frac{n_{k,t}^E}{\sum_j n_{j,t}^E}$$

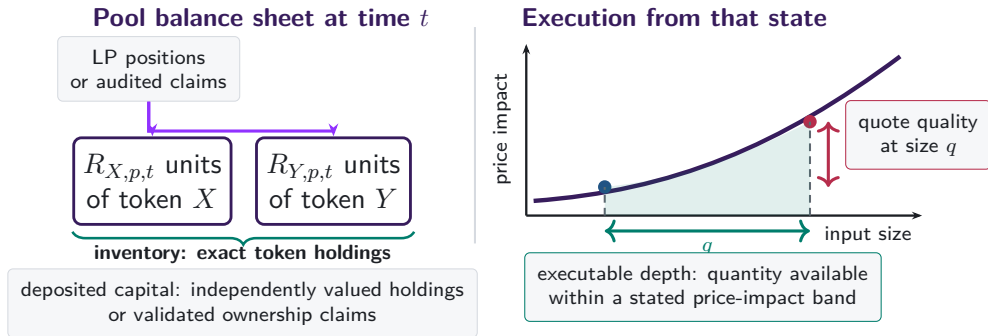
How often asset  $k$  appears at either end of the observed pool route.

The benchmark describes reconstructed pool routes; an aggregator instruction can begin or end outside the connected component.

**Excess use compares intermediation with observed route-endpoint use on the same support.**

*Note:* For USDT in 2024–26, within-category changes account for 93.7% of the count change and 85.6% of the routed-value change; reweighting between single- and cross-venue categories accounts for 6.3% and 14.4%. The identity carries no sampling interval.

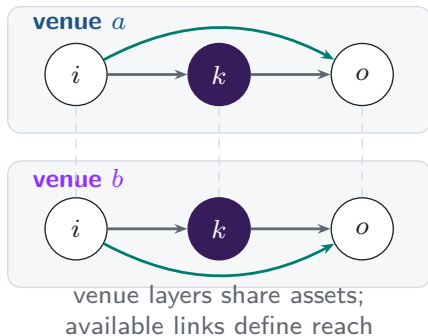
## A10. Capital, inventory, and depth are different objects



Capital is a stock. Depth and quotes depend on trade size and current pool state.

## A11. Additional venues expand the feasible route set

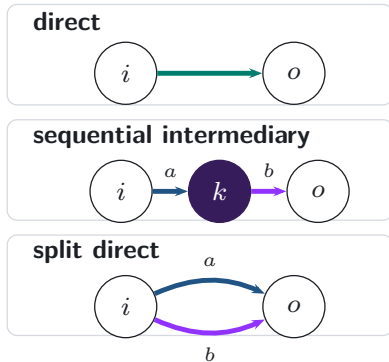
### Feasible venue layers



realised  
execution

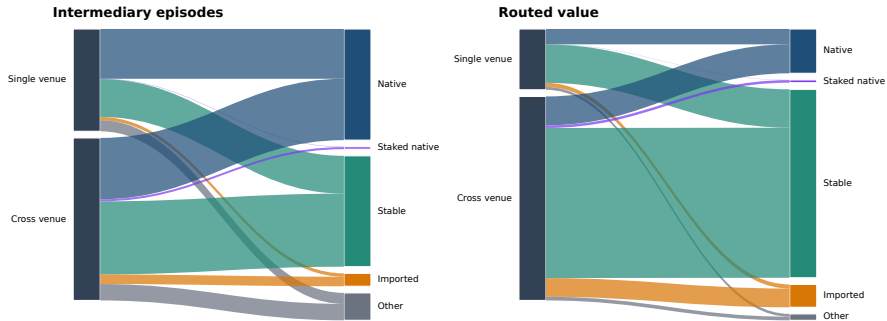


### Realised route topology



## A11b. Venue scope and vehicle type differ in 2026

**Integration and vehicle composition are separate margins in 2026**



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

*Note:* Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

## A12. References: vehicle currencies and market structure

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- Gopinath and Stein (2021), *Quarterly Journal of Economics*
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## A13. References: decentralised exchange design

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